Proposal Cover

Project Proposal for Students' Learning Task Management System "LearnTrack"

COURSE NAME: Introduction to Software Engineering

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Major: Artificial Intelligence, Group 3

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To design and implement a LearnTrack — Personal Learning Task Management Prototype

Problem:

Nowadays,university students seem to suffer from the lack of time,but it isn't totally their fault.Students receive assignment deadlines scattered across different courses,group chats and platforms and without a single place to view all upcoming tasks,so small tasks are easily forgotten until the deadline is imminent.Students do make plans,but without progress tracking and prioritisation,a daily plan is quickly abandoned—A large amount of research data indicates that 73.3% consider time management very important,yet only 32.19% actually manage their time regularly.

At the same time,long-term assignments such as term papers usually only have one distant deadline,which means students are in lack of a way to break them into weekly sub-tasks and monitor progress,resulting in last-minute all-nighters,lower quality and anxiety.

University students are willing to make study plans,but they lack a lightweight tool that enables centralized viewing of all task deadlines,decomposition of long-term assignments, progress visualization,and priority reminders.

Solution:

We propose the development of the LearnTrack personal learning-task management prototype that will deliver student-oriented academic assistance features tailored to university study rhythms.It optimizes functions for specific scenarios including coursework assignments,exam revision and group projects.Compared with general-purpose office tools,this system better adapts to students' usage habits.This enables learners to organize their academic work with reduced operating overhead

through a dedicated prototype that improves the precision and efficiency of personal study management.

Objectives:

With this project,we intend to achieve the following:

(a) design and implement the LearnTrack prototype for personal learning-task management by [date].

(b) realize core functions including task deadline viewing,long-term task decomposition,progress display and priority reminder tailored for us university students.

(c) ensure the prototype is lightweight and student-friendly to reduce operation cost and improve academic task management efficiency throughout the project period.

Benefits:

The proposed LearnTrack personal learning-task management prototype helps university students resolve common academic troubles including disorganized tasks,missed deadlines and improper time arrangement.By providing categorized task recording,deadline alerts and progress tracking tailored for study scenarios,students can replace scattered note-taking tools and achieve systematic academic planning.This lightweight tool lowers operation barriers,improves learning efficiency and cultivates good time-management habits for undergraduate users.Meanwhile, the full-cycle development process offers the project team valuable hands-on practice in software design,coding and teamwork,and the finished prototype can be reused as practical auxiliary material for student academic management.

Timeline:

The table below presents the estimated timeline of our LearnTrack personal learning-task management prototype project,which is divided into four sequential phases covering proposal drafting,prototype coding,final report writing and presentation preparation.

Phase	Description	End date
Phase 1	Proposal preparation:Write proposal sections including Problems,Solutions,Objectives and Benefits;conduct requirement research and tool investigation;sort group member information;unify document format and submit the final proposal.	Week 2
Phase 2	Prototype development and PoC implementation Complete system architecture design,Task class and JSON storage module development;implement TaskManager CRUD,filtering and statistics functions;build CLI menu and input validation;carry out unit & integration testing;write user manual,fix bugs and package the prototype.	Week 5
Phase 3	Final report writing:Finish report outline and chapter assignment; complete requirement analysis,UML diagrams,system design,implementation,testing and evaluation,conclusion and references;check page count and format consistency,submit the final report.	Week 8
Phase 4	Project presentation preparation:Deliver periodic project updates;design presentation slides and PoC demo script;record group presentation video;package all presentation files for submission.	Before the deadline

Action plan:

Objective 1:To complete and submit the project Proposal

Action	Assigned to	Deadline
Write Problem,Solution,Objectives sections	Jialiang Cheng	Week 1
Draft Benefits and collect 2–3 reference materials	Jiayu Yang	Week 1
Draft Action Plan	Zhaoliang Zhuang	Week 1
Collect 3 pain points of university students' academic task management	Xingyan Xu	Week 1
Investigate 2 task management tools	Haorong Li	Week 1
Compile potential risk list	Zekun He	Week 1
Sort out names,student IDs and majors of 9 group members	Senyu Xu	Week 1
Create Proposal submission checklist	Wenjie Shang	Week 1
Extract reference sentence templates from official template	Zixuan Deng	Week 1
Aggregate materials and preliminary format screening	Jiayu Yang	Week 2
Design cover page and format group information	Jiayu Yang	Week 2
Review,polish and unify the full document	Jialiang Cheng	Week 2
Format check,export PDF and submit	Jialiang Cheng	1 day before deadline

Objective 2:To complete core functions of the PoC prototype

Action	Assigned to	Deadline
Requirement confirmation and layered architecture design	Jialiang Cheng,Zhaoliang Zhuang	Week 3
Implement Task class	Zhaoliang Zhuang	Week 3
Implement read & write for JsonStorage	Jiayu Yang,Senyu Xu	Week 3
Implement CRUD,filtering and statistics in TaskManager	Zhaoliang Zhuang,Zekun He	Week 4
Build CLI menu and input validation	Haorong Li,Xingyan Xu	Week 4
Unit testing and integration testing	Wenjie Shang,Zixuan Deng	Week 5
Write a page user manual(PDF)	Jiayu Yang,Xingyan Xu	Week 5
Integrate bug fixes and package source code into ZIP	Jialiang Cheng,Zhaoliang Zhuang	Before PoC deadline

Objective 3:To complete the technical report

Action	Assigned to	Deadline
Create report outline and assign chapter tasks	Jialiang Cheng	Week 5
Write Introduction and Project Context	Jialiang Cheng	Week 6
Write Functional/Non-Functional Requirements	Zhaoliang Zhuang	Week 6
Complete Use Cases,use case descriptions and UML diagrams	Wenjie Shang,Zixuan Deng	Week 6
Write System Design and Architecture	Zhaoliang Zhuang,Zekun He	Week 7
Write Implementation chapter	Haorong Li,Senyu Xu	Week 7
Write Testing and Evaluation	Wenjie Shang,Zixuan Deng	Week 8
Write Conclusions and Future Work	Jiayu Yang	Week 8
Unify format of References	Haorong Li	Week 8
Appendix A:Group Workload Profile,collect signatures and compile	All members,compiled by Jialiang Cheng	Before report deadline
Check page count,format and consistency	Wenjie Shang,Zhaoliang Zhuang	Before report deadline

Objective 4:Project presentation and periodic project updates

Action	Assigned to	Deadline
Deliver project updates 1-4(at least 3 submissions)	Jialiang Cheng,Jiayu Yang(rotating)	According to course schedule
Prepare presentation PPT	Jiayu Yang,Zixuan Deng	1 week before presentation
Write demo script and PoC presentation workflow	Zhaoliang Zhuang,Haorong Li	1 week before presentation
Record MP4 video with all group members on camera	All members	Before presentation deadline
Package PPT and MP4 into one ZIP file	Jialiang Cheng	Before presentation deadline